

Will Hovel

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Education

University of California San Diego *B.S. Electrical Engineering, GPA 4.00 / 4.00* **Expected Class of 2028**

- ECE 35/65/100/102 – Circuit analysis, op-amp/filter design, BJT/MOSFET amplifiers, LTSpice
- ECE 45/171A/109/174 – Fourier/Laplace, root locus, compensator design, least-squares optimization
- ECE 5/15/25 – Embedded programming, C, combinational/sequential logic, Verilog labs

Professional Experience

Di Santo Research Lab *Research Engineer*

Jan. 2026 – Present

I study environmental effects on fish, designing a fish health monitor and a tool to measure fish swimming ability.

- Designed 4-layer mixed-signal ECG PCB for fish: AD8232 AFE, 100 V/V gain, 0.16–40.8 Hz passband
- Filtered with 1st-order HPF and 2nd-order Sallen-Key LPF, then 2nd-order IIR bandpass on Zephyr
- Rejected 60 Hz mains via submerged Right Leg Drive electrode and digital notch filter
- Routed continuous ground plane; added 10–22 μ F bulk decoupling for BLE transmit current spikes
- Integrated nRF52840 (Cortex-M4F) BLE streaming on a CR2032 with deep-sleep duty cycling
- Sampled at 12-bit / 100 Hz; sent 26-byte packed BLE payloads on peak detection
- Implemented real-time heart rate, ventilation rate, and cough classification on the M4F FPU
- Built MATLAB kinematics pipeline processing 250+ high-speed videos across 5 species
- Aligned midlines by least-squares rotation; FFT-interpolated 5–10 landmarks to 200 body positions
- Extracted tail-beat frequency, amplitude, wavelength, and curvature; added 3D tail-vector analysis
- Built 2-foil flapping school simulator (1–4 Hz) with control GUI for frequency and position
- **Awarded \$6,500** through David Marc Belkin Memorial Research Scholarship (April 2026)

Tools of Imagination *Summer Researcher (ECE)*

Jun. 2026 – Present

- Paid research under Profs. Karcher Morris and Robert Twomey on STEAM education hardware
- Built public documentation site and hands-on build guides for the EnVision ESP32-S3 dev board
- Prototyped a wearable Imagination-profile wristband on the dev board

Experience & Projects

Triton Droids *Electrical Lead – Humanoid Robot*

Oct. 2025 – Present

I lead electrical design for a full-scale humanoid robot being built to kick a soccer ball by January 2027.

- Designed Altium 48V power PCB for 15 motors: 15 A peak / 6 A continuous each, 30 A bus
- Brought up 3 PCBAs adding current limiting; ran continuity, power-up, and rail tests
- Isolated faults correlating schematics to layout with oscilloscope, DMM, and signal injection
- Led power, sensing, and control integration across platforms
- Designed hands-on onboarding (soldering, wiring, self-balancing robot); trained 20+ members

Motion Sense Glove *Embedded Systems Project*

Oct. 2025 – Present

- Designed ESP32-S3 KiCad PCB with 5-channel flex-sensor AFE and I2C IMU interface
- Sampled flex sensors at 12-bit / 50 Hz with 8 $\times$ oversampling for stable, low-noise readings
- Trained 5 $\rightarrow$ 16 $\rightarrow$ 12 $\rightarrow$ 5 MLP in NumPy, exported to C; live on ESP32-S3, 99.7% accuracy
- Added 5-sample majority vote with 0.6 confidence gate; gestures shown on SSD1306 OLED
- Hand-soldered and reworked SMD/through-hole components on custom boards

ECE 5 – UC San Diego *Lab & Course Projects*

Sept. – Dec. 2025

- Designed and soldered speaker PCB: schematic, layout, assembly, and functional validation
- **Placed 1st overall** in PID line-following robot competition out of 300+ students

Team Spyder (FIRST Robotics) *Programming & Electrical Lead*

Apr. 2021 – Jun. 2023

- Led programming and electrical for a World Championship FRC team; **Innovation in Control Award (2023)**

Skills

- Hardware: Altium, KiCad, 4-layer mixed-signal layout, PCBA bring-up, SMD/THT soldering, rework
- Embedded & RF: nRF52840 (Cortex-M4F), Zephyr RTOS, ESP32-S3, BLE, UART/I2C/SPI, low-power design
- Analog & Test: Op-amps, active filters, biopotential AFE, 48V power, oscilloscope, DMM, LTSpice
- Languages & Tools: C, C++, Python, MATLAB, Java, Verilog, Git
- Engineering: Real-time DSP, PID and root locus control, test procedure development, data logging